Customer Retention Analysis Full Methodology

UCI Online Retail II Cohort Retention & RFM Segmentation

Data Source

UCI Machine Learning Repository — Online Retail II. Real UK-based online gift retailer transactions, December 2009–December 2011, two year-sheets, ~1 million total rows. Built entirely in Excel + Power Query.

1. Data Combination

Loaded both year-sheets into Power Query. Used **Append Queries** to stack them into one combined table. The two individual year-queries (Year 2009-2010, Year 2010-2011) remain as live source queries kept intact, not deleted, so the workbook can refresh cleanly if new data is added later.

2. Flags

Invoice Type:separates real sales from cancellations, based on the UCI-documented "C" invoice prefix:

if Text.StartsWith([Invoice No], "C") then "Cancellation" else "Sale"

Required setting Invoice No to Text type first (errors otherwise).

Stock Type : separates real products from non-product entries (postage, discounts, fees, manual adjustments):

if Text.Length(Text.Select([StockCode], {"0".."9"})) < 5 then "Non-Product" else "Product"

Logic: strips everything except digits from StockCode, counts what remains. Real product codes carry 5+ digits (even with a letter suffix, e.g. 90200A); non-product codes (POST, D, BANK, TESTO, etc.) don't. Verified — not assumed — by sampling Description values against flagged rows before trusting the rule.

3. Cleaning

- **Removed ~243,000 rows with null Customer ID** unattributable to any customer, excluded for customer-level analysis specifically.
- **Trimmed whitespace** on StockCode, Customer ID, and Description caught twice letting supposedly-flagged junk values (e.g. "D " vs "D") slip past the Stock Type rule.
- **Revenue column:**

Quantity * UnitPrice

Works uniformly since signs are pre-baked (Sale positive; Cancellation and Discount rows negative).

4. Cohort Construction

Customer First Purchase: Referenced main table, trimmed to Customer ID + InvoiceDate, grouped by Customer ID, Min(InvoiceDate) → First Purchase Date.

Customer Last Purchase Date: Same pattern, Max(InvoiceDate) → LastPurchaseDate. Merged with First Purchase table on Customer ID, Inner join.

Cohort Month:

Date.ToText([First Purchase Date], "yyyy-MM")

Relationship: Cohort Month + First Purchase Date + Last Purchase Date merged back into the main transaction table, matched on Customer ID, **Right Outer join** (kept every transaction row).

Months Since First Purchase:

Date.Year([InvoiceDate]) * 12 + Date.Month([InvoiceDate])

- (Date.Year([First Purchase Date]) * 12 + Date.Month([First Purchase Date]))

Converts both dates into a running month-count and subtracts — always ≥0 since First Purchase Date is always the earliest. This is what allows cohorts joining at different calendar times to be compared fairly, at the same relative stage of their lifecycle.

5. Cohort Retention Table

- Referenced the main table
- Filtered Stock Type = Product AND Invoice Type = Sale (isolated to this reference query specifically an earlier attempt applied the filter to the main table itself, affecting every downstream query; corrected)
- Trimmed to Customer ID, Cohort Month, Months Since First Purchase
- Grouped by Cohort Month + Months Since First Purchase, **Count Distinct Rows** → Active Customers (valid because Customer ID was the only remaining column after trimming)
- Pivoted: Rows = Cohort Month, Columns = Months Since First Purchase, Values = Active Customers, shown as **"% Of," Base Field = Months Since First Purchase, Base Item = 0** (true retention rate each cell divided by that row's own Period-0 value)
- Color-scale conditional formatting applied for the heatmap

Fair cohort comparison: top cohorts compared at equivalent Months Since First Purchase stages, not lifetime totals. **2011-12 excluded** join count of 28 vs. hundreds elsewhere, confirmed as incomplete data capture.

6. RFM

Frequency: Referenced main table, filtered Product+Sale, trimmed to Customer ID + Invoice No only, grouped by Customer ID, **Count Distinct Rows** → Frequency. (First attempt kept extra columns before grouping, inflating the count fixed by trimming first.)

Monetary: Referenced main table, filtered Product+Sale, trimmed to Customer ID + Revenue, grouped by Customer ID, Sum(Revenue) → Monetary.

Last Purchase Date *(Recency-specific)*: rebuilt with the Product+Sale filter applied the original First/Last Purchase table did not have this filter and was not reused here.

Max Date: single latest InvoiceDate across the whole dataset, unfiltered stands in for "today."

Recency:

Duration.Days([MaxDate] - [Last Purchase Date])

RFM: Frequency + Monetary + Recency merged into one customer level table, matched on Customer ID, **Inner join** at each step. Row count checked after every merge to confirm no fan-out.

7. Quintile Scoring

For each of Recency, Frequency, Monetary:

1. Sort by that metric (ascending for Frequency/Monetary; **descending** for Recency)
2. Add a named Index column (M_Index, F_Index, R_Index)
3. Row count read from the preview pane and hardcoded (an earlier attempt used a live Table.RowCount() inside the formula, triggering a runaway recalculation)
4.

Number.RoundUp(([Index] + 1) / TotalRowCount * 5)

→ M_Score, F_Score, R_Score

Bug caught and fixed: Recency was initially sorted ascending instead of descending, silently inverting every R_Score. Caught because it produced a logically impossible pattern (brand-new cohorts appearing as "Hibernating/Lost"). All downstream Segment/RFM outputs were rebuilt after the fix.

8. Segment

if [R_Score] >= 4 and [F_Score] >= 4 then "Champions"

else if [R_Score] >= 4 and [F_Score] >= 2 then "Loyal Customers"

else if [R_Score] >= 4 and [F_Score] < 2 then "New/Promising"

else if [R_Score] >= 2 and [R_Score] <= 3 and [F_Score] >= 3 then "At Risk"

else if [R_Score] <= 2 and [F_Score] >= 3 then "Can't Lose Them"

else "Hibernating/Lost"

Monetary deliberately excluded from segment-defining logic kept as secondary context only.

9. Segment × Cohort

Cohort Month + Customer ID: deduplicated Customer ID + Cohort Month table (Remove Duplicates applied), merged into RFM on Customer ID, **Inner join**. (An earlier attempt merged the transaction-level Cohort Month table instead, fanning ~5,083 rows out to ~1 million fixed by deduplicating first; row count checks after every merge became standing practice from this point on.)

Pivoted with **Cohort in Rows, Segment in Columns**, Values = Sum of Monetary, shown as **% of Column Total**.

Dropped approach, noted for transparency: Segment was also merged onto the Months Since First Purchase transaction-level table, intending to show segment evolving over a customer's lifecycle. Abandoned Segment is a fixed, lifetime "final verdict" per customer, not a value that changes month to month, so the chart showed no real evolution. Replaced with a simple Segment × Customer Count summary built directly off RFM.

10. Supplementary Metrics

Gross Revenue: Sum(Revenue), filtered Stock Type = Product, Invoice Type = Sale only.

Net Revenue: Sum(Revenue), filtered Stock Type = Product, no Invoice Type filter (Sale + Cancellation combined, netting automatically).

Cancellation Rate: Referenced main table, filtered Stock Type = Product, trimmed to Invoice No + Invoice Type, **Remove Duplicates on Invoice No** (since one invoice spans multiple product lines), grouped by Invoice Type, Count Rows → Cancellation ÷ (Sale + Cancellation).

Purchase Type / Customer Count: On the Frequency table: if [Frequency] > 1 then "Repeat" else "One-Time", grouped, Repeat ÷ Total → Repeat Purchase Rate (72.34%).

Relationship Summary (all merges, build order)

Merge	Tables	Key	Join Type
1	Customer First Purchase + Customer Last Purchase Date	Customer ID	Inner
2	Main table + Cohort Month/First/Last Purchase table	Customer ID	Right Outer
3	Frequency + Monetary	Customer ID	Inner
4	(Frequency+Monetary) + Last Purchase Date (Recency)	Customer ID	Inner
5	RFM + Cohort Month + Customer ID (deduplicated)	Customer ID	Inner

Every merge was followed by a row-count check against the pre-merge table to catch unintended fan-out this caught the ~200x duplication in Merge 5's first attempt.

Query retention note: all source and transformation queries were kept intact (not deleted) to preserve refreshability only Net Revenue, Gross Revenue, and Cancellation Rate are true terminal outputs with nothing downstream depending on them, and even those were kept so the workbook continues to show up-to-date summary stats automatically if new data is ever added.